

Eduvos Assessment Coversheet

Academic Details

Campus: **MIDRAND CAMPUS**

Faculty: **INFORMATION AND TECHNOLOGY**

Module Code: **ITMDA3-34**

Lecturer's Name: **SIYANDA MBATHA**

Group (if applicable) **GROUP 9**

Assessment Details

Assessment Type: **FORMATIVE ASSESSMENT- DELIVERABLE 1**

Assignment/Brief Number: _____

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Student Details (Group, if applicable)

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Student Details (Individual, if applicable)

Student Full Name: NSUKU PABALLO MASWANGANYI

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Academic Integrity

AI Disclosure Completed



Yes



No

Declaration:

I declare that this assessment is my own original work except for source material explicitly acknowledged. I confirm that this work has not previously been submitted for assessments in any course. I acknowledge the institutional AI policy and confirm that AI tools have only been used as stipulated and documented in the AI Disclosure Appendix.

Student Signature



Date

29 JULY 2026

Lecturer Use Only

Lecturer Comments:

Marks Awarded (%):

Lecturer Signature:

Date:

Assessment

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AI-Powered Student Financial Document Management Platform

1. Introduction

Every Eduvos student who receives financial aid ends up managing a small mountain of paperwork over the course of their degree: bursary agreements, award letters, fee statements, appeal correspondence, bank confirmations. Most of it lives scattered across email inboxes, chat apps and phone galleries, and very little of it gets read closely until a deadline is missed or a query comes in from the funder.

Folio was built to close that gap. It is a secure, cloud-native platform, accessible on web and mobile, that lets a student upload their financial documents once and then interact with them intelligently: searching them in plain language, asking questions about specific conditions, and getting AI-generated summaries grounded directly in the documents themselves rather than in generic advice.

This chapter sets out the reasoning behind the project: the background that shaped it, the aim it is working toward, the problem it responds to, who benefits from it and how, the boundaries deliberately placed around its scope, and the features that separate it from the document tools students already have access to.

2. Background of the Research

The shift from simple digital storage to AI-assisted document handling has been rapid. Traditional systems could store a file and let a user search it by keyword; they could not tell a student what a bursary clause actually meant. That changed with the arrival of large language models capable of summarisation, classification and question answering, and more specifically with Retrieval-Augmented Generation (RAG), which grounds an AI's answer in a user's own retrieved documents rather than in whatever the model happened to learn during training. Financial documents are full of specific figures, dates and conditions, so that grounding matters a great deal. A generic answer about how bursaries usually work is far less useful to a student than an answer pulled directly from their own award letter.

Financial-domain language models such as FinBERT show that this kind of domain-specific understanding is achievable in practice, extracting monetary values, obligations and terms from text with far more precision than a general-purpose model (Kim and Yang, 2024). At the same time, natural-language interfaces have matured to the point where users no longer need to phrase a request as a keyword search; conversational and intent-based search systems can now interpret what a user actually wants and

return the right result directly (Mo et al., 2025). Folio's AI chatbot and search bar are built on exactly this shift.

None of this is possible at scale without cloud infrastructure. Services such as AWS provide the storage, compute and security controls needed to run a document-intensive platform without the cost of managing physical servers, and their well-architected guidance around security and reliability shaped several of Folio's own design decisions (Amazon Web Services, n.d.). Combined with a mobile-first delivery model, since most students reach for their phone before their laptop, this made a cross-platform, cloud-native build the obvious direction for the project.

3. Aim of the Research

The aim of this study is to design, build and evaluate Folio: a secure, cloud-native AI platform that helps Eduvos students store, understand and act on their financial documents, and to test whether the platform can be sustained financially through a credit-based AI usage model without shutting out the students who need it most.

Following a Design Science Research approach, the project treats the platform itself as the primary research output rather than a supporting illustration (Hevner et al., 2004). The working prototype is not incidental to the study; it is the artefact being evaluated. More specifically, the study aims to:

- determine how AI and RAG can be applied to extract, summarise and answer questions about student financial documents;
- design and implement a secure architecture (authentication, encryption, audit logging) appropriate for highly sensitive personal financial data;
- build and test a natural-language search interface capable of interpreting intent-based, rather than keyword-based, student queries; and
- design and evaluate a credit-based monetisation model for the AI chatbot that balances platform sustainability against student affordability.

4. Problem Setting

Eduvos students receiving institutional bursaries, scholarships or student loans deal with a steady stream of financial documents throughout their studies, many carrying real consequences if missed: renewal deadlines, appeal windows, changes in funding conditions. There is currently no dedicated tool built around that specific problem. Cloud storage apps such as Google Drive or Dropbox will hold the files, but they do not read them. The Eduvos student portal provides access to some documents but was not

designed as a mobile-first, AI-assisted experience, and it does not answer questions about what a document actually says.

There is also a harder technical problem underneath this. Any system handling student financial data has to treat security as a starting requirement, not something added later: this is personal, sensitive information, and its misuse would cause genuine harm. Under South Africa's Protection of Personal Information Act, any platform processing this kind of data carries direct legal obligations around consent, data minimisation and safeguarding (Republic of South Africa, 2013). A prototype that ignores this is not just incomplete; it is non-compliant.

5. Benefits of the Study

For Eduvos students, the most direct benefit is time and clarity: a single place to store financial documents, and an AI layer that can answer questions such as when a bursary renewal is due, or what happens if a payment is missed, without a student having to re-read a five-page agreement under pressure. Students who might otherwise miss a deadline buried in dense funding-agreement language get a second, AI-assisted line of defence.

Academically, the study contributes a working example of how AI adoption factors, particularly perceived usefulness and ease of use (Davis, 1989), play out in a real, deployed prototype rather than a purely theoretical model, in a domain that has had very little dedicated research attention so far.

6. Delimitation of the Study

This study is scoped deliberately. It does not attempt to:

- replace official Eduvos or other institutional financial systems: Folio manages a student's own document copies and their understanding of them, not the underlying funding transactions;
- process real student financial data during the research and prototyping phase: all prototype demonstrations use synthetic or anonymised sample documents;
- implement a production-grade payment sandbox for the credit-based monetisation model: the credit system is designed and demonstrated conceptually within the prototype rather than connected to live billing infrastructure;
- generalise findings beyond the Eduvos student population, given the sample size and institutional focus of the study; or

- provide financial or legal advice: AI-generated outputs remain informational aids with source citations, not authoritative determinations.

7. Distinguishing Features

An AI chatbot grounded in the student's own documents. Rather than answering from general knowledge, Folio's chatbot uses a RAG pipeline to retrieve the relevant passage from a student's uploaded files before responding. A student could ask something like “draft a summary of what I need to know for my interview according to this file” and receive an answer built from the actual document, with the source passage cited, not a generic interview-prep script.

A natural-language search bar. Instead of keyword search across folders, Folio's search bar is built to interpret intent. A query such as “show me the files I need to apply for a job as a software engineering graduate” is parsed for meaning rather than matched literally, drawing on the same advances in conversational, intent-based search that have reshaped information retrieval more broadly (Mo et al., 2025).

A credit-based monetisation model. To keep the platform financially sustainable without an open-ended AI usage cost, each student account includes a set number of free AI chatbot questions per month. Once that allowance is used, students can purchase additional credits to keep asking questions. This mirrors the credit-based pricing structures that have become common across AI-powered SaaS products more broadly, where usage-based and credit-based models let providers absorb variable inference costs while giving users predictable, prepaid access (McKinsey & Company, 2025; Bessemer Venture Partners, 2026). Document storage, upload and the NLP search bar remain unrestricted regardless of credit balance; the limit applies only to AI chatbot queries, which carry the highest processing cost.

Security built in from the identity layer up. Authentication runs through OAuth 2.0 and OpenID Connect via Eduvos Single Sign-On, so students never need a separate password for their most sensitive documents (Hardt, 2012; OpenID Foundation, 2023).

References

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